

CSC311 Introduction to Machine Learning Week 7 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W7-T1. The bias-variance decomposition decomposes generalization error into three terms. What are these three sources of error? For each source, define and explain that source of error.

W7-T2. We saw the following bias-variance decomposition for a regression example, using the square loss:

$$\mathbb{E}[(y - t)^2] = \mathbb{E}[(\bar{y} - y_\star)^2] + \text{Var}(y) + \text{Var}(t)$$

where $y = \hat{f}(\mathbf{x})$

- (a) What does the $\mathbb{E}[\cdot]$ symbol mean? That is, an expectation is always computed with respect to some random variable and its distribution. What is the random variable, and what distribution is it sampled from?
- (b) What does $\hat{f}$ represent? How is it generated? (How does this question relate to the previous question?)
- (c) What do y and $y_\star$ represent?
- (d) What does the term $\mathbb{E}[(\bar{y} - y_\star)^2]$ represent? Explain.
- (e) What does the term $\text{Var}(y)$ represent? Explain.
- (f) What does the term $\text{Var}(t) = \mathbb{E}[(y_\star - t)^2]$ represent? Explain.

W7-T3. Which of the following statements are true about the bias-variance decomposition?

- (a) A model that is high bias tends to overfit.
- (b) Reducing the number of features used to train a model increases the Bayes error.
- (c) Consider a linear model $\mathbf{y} = \mathbf{w}^T \mathbf{x}$ and a training data set $\mathcal{D}$. The “variance” of this model can be estimated by training this model using $\mathcal{D}$, using this model to perform inference on a test set $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}$ to produce estimates $y^{(1)}, y^{(2)}, \dots, y^{(n)}$, and estimating the sample variance $\frac{1}{n} \sum_{i=1}^n (y^{(i)} - y_{\text{mean}})^2$, where $y_{\text{mean}} = \frac{1}{n} \sum_{i=1}^n y^{(i)}$.
- (d) Increasing k in a k -nearest neighbour model increases the variance of the model.
- (e) Increasing the max depth of a Decision Tree model increases the variance of the model.
- (f) Increasing the regularization parameter λ in a neural network increases the variance of the model.
- (g) Increasing the number of hidden units in a 2-layer neural network increases the variance of the model.
- (h) The number of training data points increases the variance of the model.

W7-T4. Which of the following statements are true about bagging?

- (a) Bagging involves training the same classifier on resampled versions of the training set.
- (b) The bagged model has higher bias than a single model.
- (c) The bagged model has lower variance than a single model.
- (d) Reducing the correlation between models in the ensemble can reduce the variance of the bagged model.
- (e) A random forest classifier uses bagging.
- (f) Bagging is more computationally expensive than using a single model.

W7-T5. Suppose you flip a coin $N = 100$ times and get outcomes $\mathcal{D} = \{x_1, \dots, x_N\}$ where $x_i \in \{0, 1\}$ and $x_i = 1$ is interpreted as heads H . The coin is possibly loaded, and we would like to estimate the probability of obtaining heads in future coin flips.

- What distribution would you use to model this probabilistic system? What parameter does this distribution have?
- Write down the likelihood $p(\mathcal{D})$ of the data. Note down, in your derivation, where you use each of the two assumptions: the i.i.d. assumption, and the distributional assumption from part (a).
- Write down the log likelihood $\log p(\mathcal{D})$ of the data.
- Produce an estimate for your parameter by maximizing the log likelihood. Show all your work.

Validation

W7-V1. Suppose that you have the following data set, and we would like to estimate $P(Y|X_1, X_2)$

X_1 (ChestPain)	X_2 (Smoking)	Y (Heart Disease)	Count
1	1	0	10
1	1	1	250
1	0	0	40
1	0	1	50
0	1	0	30
0	1	1	200
0	0	0	5
0	0	1	415

- Suppose we wish to model $P(Y|X_1, X_2)$ directly, as we did in class. Thus, each $P(Y|X_1 = 1, X_2 = 1)$, $P(Y|X_1 = 1, X_2 = 0)$, $\dots$, will be a separate distribution to be learned. List all the parameters in this model that will need to be learned.
- Suppose we wish also wish to compute/model $P(Y)$ directly. Explain why this can be done by modeling $P(X_1, X_2)$. How many parameters are required in this distribution?
- Use maximum likelihood to estimate all the parameters above.

W7-V2. Suppose again that you would like to build a model using the data set from the previous question in order to predict heart disease. Suppose instead that you decide to model $P(X_1, X_2|Y)$ instead.

- How would you be able to compute $P(Y|X_1, X_2)$ in terms of $P(X_1, X_2|Y)$? What other terms would you need to be able to compute?
- You should see a term $P(X_1, X_2)$ above. How can you compute this in terms of the other quantities from part (a)?
- Why will we need to model $P(Y)$? What distribution could be used to model Y , and what parameter does it have?
- What distribution does each of $P(X_1, X_2|Y = 1)$ and $P(X_1, X_2|Y = 0)$ have? And what parameters?
- How many parameters in total need to be learned? How does this compare with the number of parameters learned in the previous question?
- Use maximum likelihood to estimate all the parameters above.

Generalization

W7-G1. Suppose we have the dataset from class, where there is a single continuous feature X (e.g., Cholesterol) that will be used to predict the presence of heart disease. That is, we want to use prior data to answer questions like "What is the probability that a new patient who has blood cholesterol of 4.51 has heart

disease?” We discussed in class that probabilistic reasoning could be used for this problem. (We will discuss this example in depth in week 10, but for now, as a generalization exercise...)

X (Cholesterol)	Y (Heart Disease)
3.9	0
5.1	0
5.4	0
4.5	1
6.5	1
8.7	0
9.3	1

- (a) We will model $P(Y)$ as a Bernoulli distribution. What parameter would need to be learned?
- (b) We will model $P(X|Y = 1)$ and $P(X|Y = 0)$ using normal distributions. What parameters would need to be learned?
- (c) Write down the log-likelihood for the above data set and for this distribution. We recommend not plugging in the data right away. Instead, show that the log-likelihood “decomposes”, and that the parameters for $P(Y)$, $P(X|Y = 1)$ and $P(X|Y = 0)$ can be learned separately.
- (d) Estimate the parameter for $P(Y)$ using maximum likelihood.
- (e) Estimate the remaining parameters using maximum likelihood. You might need to read ahead a bit to see how.
- (f) Estimate $P(Y = 1|X = 4.51)$ using your learned model.